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EVOLUTIONARY REVERSALS OF ANTIBIOTIC RESISTANCE IN EXPERIMENTAL POPULATIONS OF *PSEUDOMONAS AERUGINOSA*

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Antibiotic resistance mutations are accompanied by a fitness cost, and two mechanisms allow bacteria to adapt to this cost once antibiotic use is halted. First, it is possible for resistance to revert; second, it is possible for bacteria to adapt to the cost of resistance by compensatory mutations. Unfortunately, reversion to antibiotic sensitivity is rare, but the underlying factors that prevent reversion remain obscure. Here, we directly study the evolutionary dynamics of reversion by experimentally mimicking reversion mutations—sensitives—in populations of rifampicin-resistant *Pseudomonas aeruginosa*. We show that, in our populations, most sensitives are lost due to genetic drift when they are rare. However, clonal interference from lineages carrying compensatory mutations causes a dramatic increase in the time to fixation of sensitives that escape genetic drift, and mutations surpassing the sensitives' fitness are capable of driving transiently common sensitive lineages to extinction. Crucially, we show that the constraints on reversion arising from clonal interference are determined by the potential for compensatory adaptation of the resistant population. Although the cost of resistance provides the incentive for reversion, our study demonstrates that both the cost of resistance and the intrinsic evolvability of resistant populations interact to determine the rate and likelihood of reversion.

KEY WORDS: Compensatory evolution, evolvability, genetic drift, probability of fixation, rifampicin.

Antibiotic resistance mutations are usually accompanied by a fitness cost, expressed in terms of reduced competitive ability, transmission rate, and virulence relative to sensitive strains (Anderson and Levin 1999; Andersson 2006). There are two possible evolutionary responses to this cost of resistance when populations of resistant bacteria encounter an antibiotic-free environment, as occurs during transmission between hosts or when antibiotic use is halted. First, bacterial populations can eliminate the cost of resistance by reverting to antibiotic sensitivity. Second, bacterial populations can adapt to the cost of resistance by fixing second site compensatory mutations that fully or partially recover the cost of resistance without compromising antibiotic tolerance (Levin et al. 2000; Reynolds 2000; Nagaev et al. 2001; Arason et al. 2002; Maisnier-Patin et al. 2002; Comas et al. 2012). Unfortunately, compensation occurs much more commonly than rever-

sion, both in vivo and in vitro (Andersson 2003; Anderson and Hughes 2010; Andersson and Hughes 2011), but the underlying population genetic reasons for this are not completely understood.

Three key factors likely constrain reversion. First, reversion mutations are rare relative to compensatory mutations; it has been estimated that there are an average of 11–12 compensatory mutations per resistance mutation (Poon et al. 2005), although some may have as many as 35 or more (Björkman et al. 2000). The abundance of compensatory mutations is likely to impose an especially important constraint on reversion because compensatory mutations show epistatic interactions with resistance mutations, such that reversion mutations incur a fitness cost once bacterial populations have acquired compensatory mutations. Second, genetic drift will lead to the stochastic extinction of reversion mutations when they are at a low frequency. Finally, competition

between reversion mutations and compensatory mutations (i.e., clonal interference) may constrain reversion via two mechanisms. Compensatory mutations that do not fully recover the cost of resistance are expected to delay, but not prevent, the fixation of reversion mutations (De Visser and Rozen 2006). Mutations that recover fitness to higher than wild-type levels, however, are expected to prevent the fixation of reversion mutations (Gerrish and Lenski 1998). Here, we define mutations in the resistant background that surpass the current fitness of the sensitive type as “super-compensatory” mutations (note that this does not preclude adaptation in the sensitive type, only that the rate of such adaptation is lower than the rate of compensatory adaptation). Although it is clear that the relative rarity of reversion mutations helps contribute to the maintenance of antibiotic resistance, the impact of drift and clonal interference on reversion has not been investigated.

The goal of this article is to investigate how drift and clonal interference affect the evolutionary dynamics of reversion using an experimental model system based on the evolution of rifampicin resistance in *Pseudomonas aeruginosa*. Rifampicin is a potent broad-spectrum antibiotic that binds to a highly conserved domain on the β -subunit of RNA polymerase (*rpoB*), preventing the elongation of RNA transcripts. Resistance to rifampicin in many bacteria, including *P. aeruginosa*, results from mutations in the rifampicin binding domain that compromise the ability of rifampicin to bind to its target (Wehrli 1983). Rifampicin resistance mutations in *P. aeruginosa* tend to impart a fitness cost in the absence of antibiotics, most likely resulting from compromised RNA polymerase function (Hall et al. 2011). Previous work has shown that adaptation to the cost of rifampicin resistance in *P. aeruginosa* usually occurs by compensatory mutations, often at second sites in *rpoB*, but complete reversion to antibiotic sensitivity is also possible (Hall et al. 2010).

The conventional method that has been used to study reversion and compensation is to challenge populations of antibiotic-resistant bacteria with adapting to the cost of resistance in vitro and in vivo (reviewed in Andersson and Hughes 2010). Following some period of selection, the fitness and resistance of evolved populations are then determined relative to suitable control populations to test for both compensatory adaptation and reversion. Although this approach has consistently shown that compensation tends to occur more commonly than reversion, it does not provide any insight into whether reversion is rare simply because of a paucity of reversion mutations or because of a low probability of fixation of reversion mutations due to drift and clonal interference.

To study reversion, we used an experimental model system in which populations of two rifampicin-resistant strains of *P. aeruginosa* were seeded with individual cells of a genetically marked rifampicin-sensitive strain, effectively mimicking a spontaneous

reversion mutation. We then followed the evolutionary dynamics of sensitive and resistant lineages over approximately 300 generations in a highly replicated selection experiment. This experimental approach has two important benefits. First, because we seeded populations with a sensitive strain, we could be confident that any lack of reversion was not simply attributable to a low rate of supply of reversion mutations. Second, because the sensitive that we introduced was genetically marked, we were able to accurately follow the dynamics of sensitive lineages, even when sensitives were extremely rare (frequency of less than 1 in 2×10^5). Using this approach, we showed that despite the large cost of resistance, most sensitives were lost due to genetic drift when they were initially rare. The main effect of clonal interference was to dramatically increase the time to fixation of reversion mutations, although in some cases reversion mutations that escaped genetic drift were ultimately eliminated by competing supercompensatory mutations. Crucially, we found that the ultimate probability of fixation and time to fixation, following drift and clonal interference, were determined by the intrinsic evolvability of the resistant population. As the resistant mutants we studied had virtually indistinguishable fitness costs with respect to their common sensitive ancestor, this suggests that the ability to compensate for resistance, and not the cost associated with resistance, may be more important in determining the probability of reversion. Intriguingly, we found evidence that the nature of evolvability differed in these two strains: compensatory mutations were seemingly more common in one resistant lineage whereas, supercompensatory mutations were more common in the other.

Materials and Methods

STRAINS

We used two rifampicin-resistant strains of *P. aeruginosa*, which are both mutants from the same PA01 clone. Resistant mutants were isolated on solid M9KB agar plates (10.5 g/L M9 Broth [Fluka/Sigma-Aldrich], 10 g/L glycerol, 10 g/L proteose peptone no. 3, 1 mL of 1 M MgSO_4 after autoclaving) supplemented with 62.11 $\mu\text{g/mL}$ rifampicin (previously described in MacLean and Buckling, 2009). Sequencing determined that both strains had mutations in the *rpoB* gene, one a single base pair substitution (C1607T, corresponding to amino acid change S536F) and the other a three base pair insertion (1552CGC1553, a proline insertion between 517 and 518, +P518). For convenience, we will refer to these mutants by their amino acid changes. We used rifampicin-sensitive PA01 strains marked with either green fluorescent protein and gentamicin resistance (GFP-PA01), or yellow fluorescent protein and streptomycin resistance (YFP-PA01) as the sensitive competitor strains (Klausen et al. 2003). GFP-PA01 and the two rifampicin-resistant strains were used to determine the cost of

resistance and in the selection experiments, whereas YFP-PA01 was used as a reference strain to measure competitive fitness.

PLATE READER-BASED COMPETITIVE FITNESS ASSAYS

To enable us to conduct competitive fitness assays with many strains, we developed a plate reader-based fitness assay using a fluorescent marker to quantify the proportion of the population made up by a marked competitor, versus total population density. Competitive fitness of a nonfluorescently labeled “strain of interest” was measured against YFP-PA01, standardized by the competitive fitness of wild-type PA01. Fluorescence intensity (FI) was measured using a combined fluorometer/luminometer (FLUOstar OPTIMA; BMG Labtech, Offenburg, Germany), with filters optimized for detecting the peak of the YFP emission spectrum (excitation = 500 ± 5 nm, emission = 540 ± 5 nm). Yellow fluorescent protein was chosen over GFP as the fluorescent marker for these competition assays to reduce detection of endogenous fluorescent proteins produced by *P. aeruginosa* (such as pyoverdine), which have emission spectra that overlap with GFP (Cox and Adams 1985; Tsien 1998). Total population density of each culture was measured using BacTiter-Glo™ (Promega, Madison, WI), which measures total population ATP using a bioluminescent enzyme. We mixed 25 μ L of culture with 25 μ L of prepared assay reagent and measured total luminescence intensity (LI) using the same fluorometer/luminometer. Fluorescence intensity and LI were measured relative to a standard curve (i.e., a dilution series of a pure culture of the YFP-PA01 strain), to give standardized total population density (T) and standardized labeled density (L). Proportion of the population made up by unlabeled strains U was then calculated as $U = T - L$. Fitness of the unlabeled strain relative to the labeled strain was then calculated by comparing the initial (i) and final (f) proportions of the unlabeled and labeled strains, or $w_{UL} = \log_2(U_f/U_i)/\log_2(L_f/L_i)$. Preliminary work showed that competitive fitness measured using this method correlates strongly (with slope indistinguishable from one) with fitness measured from colony counts on selective media (slope = 1.01 ± 0.04 , $F_{1,8} = 619$, $R^2 = 0.98$, $P < 0.0001$; see Fig. S1).

COST OF RESISTANCE

To measure the cost of resistance, we directly competed our resistant mutants against rifampicin-sensitive GFP-PA01 in minimal culture medium supplemented with serine as the carbon and nitrogen source (M9-serine: 10.5 g/L M9 Broth 1 mL of 1 M MgSO_4 , and 3.75 g/L [35.7 mmol/L] L-serine). Precompetition cultures were prepared by inoculating single colonies of the resistant and sensitive strains into M9KB broth and incubating these cultures overnight at 37°C with constant shaking at 200 rpm (overnight density $\sim 10^9$ cells/mL). Precompetition cultures of the resistant strains and the sensitive strain were then mixed at a 10-

to-1 volume-to-volume ratio and diluted by 10^{-4} into M9-serine to set up five replicate competition cultures that were incubated overnight at 37°C. We determined the initial and final titer of strains by plating samples of cultures before and after competition on M9KB agar supplemented with 64 mg/L rifampicin (to select for the resistant strain) and 40 mg/L gentamicin (to select for the resistant strain). We computed the relative fitness of the resistant strain ($w_{\text{resistant}}$), relative to the sensitive strain, as the log-ratio of the number of doublings. The cost of resistance is defined as the difference between the sensitive strain's fitness (standardized at 1) and the resistant strain's fitness.

INTRODUCTION OF SENSITIVE STRAIN

To study the dynamics of reversion, we inoculated a genetically marked sensitive strain (GFP-PA01) into populations of S536F and +P518. To initiate the experiment, we inoculated 384 populations of each genotype (in eight 96-well microliter plates) of 5.25×10^4 cells of S536F and +P518 into 200 μ L M9-serine. Each rifampicin-resistant population received an average of 0.37 cells of the rifampicin-sensitive GFP-PA01, which was achieved by serial diluting an overnight culture of GFP-PA01 grown in liquid M9KB. Preliminary experiments demonstrated that the number of sensitive cells per population followed a Poisson distribution, therefore 118.75 ± 9.05 SD populations were expected to receive at least one sensitive cell. (The number of populations follows a binomial distribution $B \sim [p = 1 - e^{-0.37}, n = 384]$, see online supporting information for full calculations.) A statistical determination of the number of populations “seeded” with the sensitive strain was necessary as there was no method at our disposal sensitive enough to detect the small number of sensitive cells in the presence of the rifampicin-resistant cells. By the Poisson distribution, the number of sensitive cells seeded into each population was expected to be either one ($n = 98$), two ($n = 18$), or three ($n = 3$)—no populations were expected to receive four or more sensitive cells. This represents an average initial sensitive frequency of 2.2×10^{-5} , if weighted by the distribution of sensitives. We estimated that, by chance alone, there could be a difference of approximately 10.22 successful sensitive inoculations between the two rifampicin-resistant mutants (see online supporting information), meaning that any difference in outcomes detected between S536F and +P518 must differ significantly from this amount. The 265 populations that did not receive a sensitive were selected alongside those that did to determine the rate and extent of compensation possible in these resistant genetic backgrounds.

SELECTION EXPERIMENT

All 384 populations were then transferred over 300 generations in a 24-h serial transfer protocol. From each population, 1 μ L was transferred by pin replicator to 200 μ L of fresh medium

(dilution factor $D = 0.005$). Throughout the selection experiment, final population size was on the order of 10^7 after 24 hours. A subset of populations (96 for each resistant genotype) was stored at -80°C .

Approximately every six to seven generations, we used simple phenotypic assays to test for the fixation of the sensitive (minimum detectable frequency of 1 in 2×10^5 or 0.0005%) or the resistant strain (minimum detectable frequency of 1 in 4×10^3 or 0.025%), assuming that the loss of one strain was equivalent to the fixation of the other. Although the frequencies at which we define fixation are arbitrary, they were chosen to be stringent so that our assays represent true fixations rather than simply failing to detect either strain present in the population. Presence of the GFP-PA01 rifampicin-sensitive strain was assayed by pin replicating 1 μL of the undiluted overnight culture on an M9KB agar plate with 40 mg/L gentamicin. Presence of the rifampicin-resistant strains was assayed by pin replicating 1 μL of the 1/200 dilution of the overnight culture on an M9KB agar plate with 64 mg/L rifampicin. Differences in the proportion of sensitives present in both rifampicin-resistant genetic backgrounds were assessed using a two-sample test for equality of proportions (without Yates' correction), function `prop.test()` in R version 2.14.1 (R Development Core Team 2011).

FREQUENCY OF SENSITIVE STRAIN

In a subset of five populations of each resistant genotype, we tracked the frequency of the sensitive whereas populations were polymorphic for resistance (generations 0–150). We measured the frequency of the sensitive strain and the total population size using the same method as for the plate reader-based competition assays, except we measured FI in two replicates and LI in four replicates.

COMPENSATORY ABILITY

To measure the ability of S536F and +P518 to acquire compensatory mutations, we measured the competitive fitness (against YFP-PA01) of a subset of populations that, by chance, either received no sensitive cells or where the sensitive never reached detectable frequency. Although some of this subset may have been seeded with the sensitive strain, the fact that it never reaches detectable frequency makes it unlikely to have had any effect on the adaptation of the resistant strain, and hence we treat these populations as though they were pure cultures of the resistant strain.

Results

FITNESS OF FLUORESCENTLY LABELED PA01 REFERENCE STRAINS

Using a selective plating competition assay, the two fluorescently labeled strains did not differ significantly in fitness from wild-

type PA01 (PA01 fitness vs. GFP: $w_{\text{PA01}} = 0.975 \pm 0.036$ SEM, $t_4 = -0.7$, $P > 0.2$; PA01 vs. YFP: $w_{\text{PA01}} = 0.987 \pm 0.032$, $t_4 = -0.4$, $P > 0.3$). We found no significant difference in competitive fitness when the two fluorescently labeled strains were directly competed (GFP vs. YFP: $w = 0.979 \pm 0.069$, t -test against mean of 1: $t_4 = -0.3$, $P > 0.35$). Preliminary work has shown that fitness measurements of different strains against GFP and YFP are well correlated (slope = 0.98 ± 0.03 , $F_{1,8} = 850$, $R^2 = 0.98$, $P < 0.0001$; see Fig. S2), hence GFP and YFP can be used interchangeably for measuring competitive fitness.

COST OF RIFAMPICIN RESISTANCE

To test for a fitness cost associated with resistance, we directly competed rifampicin-sensitive GFP-PA01 against two rifampicin-resistant mutants (S536F and +P518) in M9-serine culture medium. We found that in this environment, these resistance mutations were associated with large and indistinguishable fitness costs when compared to their sensitive PA01 ancestor ($w_{\text{S536F}} = 0.401 \pm 0.0269$ SE, $w_{\text{+P518}} = 0.387 \pm 0.0640$ SE, Welch's t -test: $t_{6,87} = 0.20$, $P = 0.42$).

PROBABILITY OF ESCAPING GENETIC DRIFT

Population genetic theory predicts that even highly beneficial mutations are likely to be lost from populations due to genetic drift when they are initially rare (reviewed in Patwa and Wahl 2008). In agreement with theory, we found that most sensitives fail to reach a detectable frequency: we identified sensitives above our detection threshold of 1 in 4×10^3 in 38 out of 118 populations of S536F (proportion = 0.32 ± 0.043 SE) and 54 out of 118 populations of +P518 populations (proportion = 0.46 ± 0.046 SE). We found that the number of populations containing sensitives rapidly decreased, as we would expect if rare sensitive populations were lost due to stochastic extinction (Fig. 1). The proportions of detectable sensitives in each rifampicin-resistant background differed significantly ($\chi^2_{12} = 4.56$, $P = 0.033$); however, the observed difference ($d = 54 - 38 = 16$) did not differ significantly from the expected difference in the number of wells receiving at least one sensitive ($d = 10.22$, $P = 0.211$, see online supporting information), hence we cannot conclude that there was a difference in the probability of escaping drift.

Is this high rate of loss of sensitive lineages due to drift surprising? Haldane (1927) famously predicted that the likelihood that a weakly beneficial mutation survives stochastic extinction when it is rare is equal to $2s$, for mutations with small s . In the context of fixing sensitive mutants in resistant populations, s is the fitness advantage of being sensitive, which is inversely proportional to the cost of resistance: $s = 1/w_{\text{resistant}} - 1$ (because $w_{\text{sensitive}} = 1$). For our resistant strains, s cannot be considered small and so the sensitive lineages should have been expected to survive; accounting for measurement error in the cost of resistance, s for

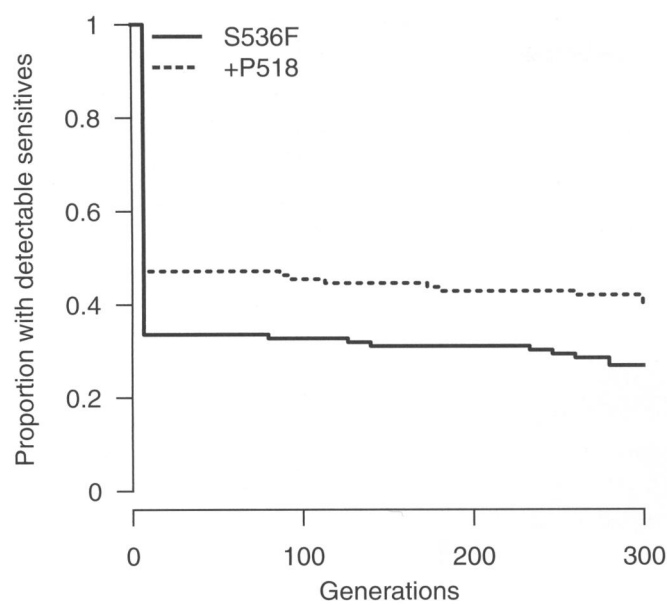


Figure 1. Survival of sensitive lineages. This figure shows the proportion of populations ($n = 118$) with detectable sensitives (sensitive frequency $> 0.0005\%$) in selection lines founded by the S536F (solid line) and +P518 strains (dotted line).

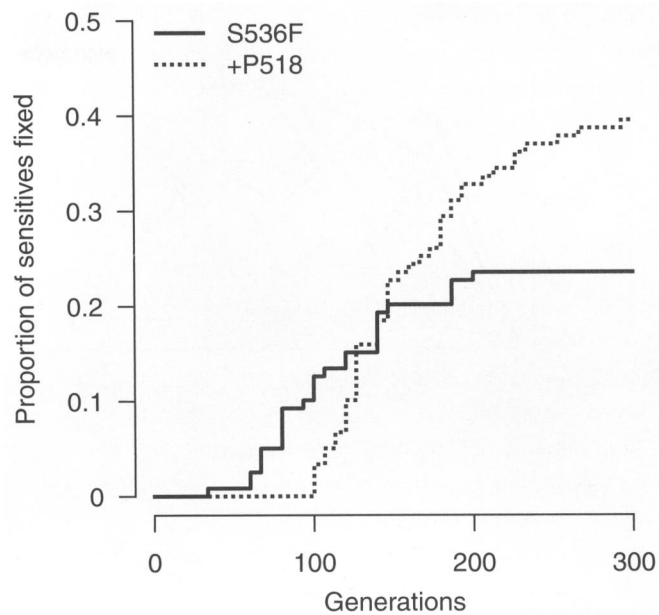


Figure 2. Fixation of sensitive lineages. This figure shows the proportion of populations ($n = 118$) where the sensitive had replaced the resistant strain (resistant frequency $< 99.975\%$) in populations founded by S536F (solid line) and +P518 (dotted line).

the sensitive strain is 1.505 ± 0.168 SE in the presence of S536F and 1.66 ± 0.439 SE in the presence of +P518. However, more recent results suggest that drift also depends on either final population size and bottleneck size (Levin et al. 2000; Wahl et al. 2002). Levin et al. (2000) showed an increase in the probability of fixation with an increase in population size and bottleneck size. Wahl et al. (2002) extends Haldane’s model assuming reproduction by binary fission and recurrent populations bottlenecks and estimates that the fixation probability of a beneficial mutation is maximized with a dilution factor of $D \approx 0.135$, more than an order of magnitude larger than our $D = 0.005$, which may have contributed to the loss of sensitives in our populations. Under this model, we can generate predictions for the probability of escaping drift that range from 0.38 to 0.47 for S536F and from 0.34 to 0.59 for +P518. These predicted ranges overlap with the 95% confidence intervals of our observed probabilities of escaping drift: from 0.24 to 0.41 for S536F and from 0.37 to 0.55 for +P518. We therefore cannot reject the predictions from this model for either genotype.

TIME TO FIXATION

In the absence of clonal interference, beneficial mutations that escape genetic drift are expected to fix rapidly as a result of natural selection. For example, the passage time between when a back mutation to sensitivity becomes detectable (i.e., frequency $> 0.0005\%$) and when it becomes fixed (i.e., frequency $> 99.975\%$) under the conditions of our experiment is expected to be 8–15

generations (Bell 2008). In contrast to this prediction, we found that sensitives that became fixed had passage times in excess of 100 generations, representing a 10-fold increase in passage time relative to our theoretical expectation (Fig. 2). Intriguingly, we found that the time to fixation differed significantly between resistance mutations (Welch’s t -test: $t_{59.8} = -4.32$, $P < 0.0001$), with a mean time to fixation of sensitives in 110.8 ± 8.4 SE generations for S536F and 157.8 ± 6.9 SE generations for +P518.

One potential explanation for the large discrepancy between expected and observed time to fixation is that clonal interference slows the fixation of the sensitive strain by effectively reducing the selection coefficient associated with sensitivity. To test this possibility, we followed the dynamics of reversion mutations in five populations of each resistant genotype where sensitives eventually became fixed and calculated the selection coefficient associated with the reversion mutations. Sensitives initially increased in frequency from 0.002% to about 50% within approximately 30 generations, followed by a slow approach to a frequency of 100% over the next 120 generations (Fig. 3A, B). In terms of selection coefficients, this represents a decline in the selection coefficient associated with reversion mutations of $s = 1.5$ at the outset of the experiment to approximately $s = 0.01$ during the following 250 generations (Fig. 3 C).

LIKELIHOOD OF FIXATION

Surviving genetic drift does not guarantee that a beneficial mutation will be fixed by natural selection in an asexual population

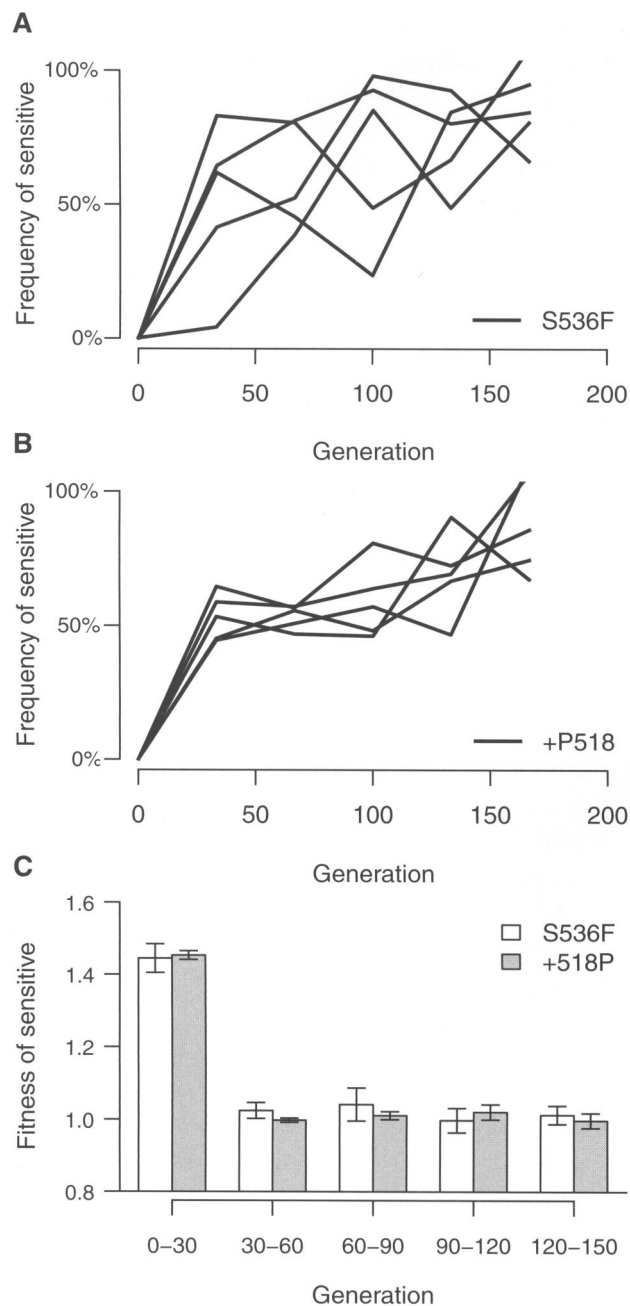


Figure 3. Dynamics of reversion. This figures shows the frequency of randomly chosen sensitives that were eventually fixed by selection ($n = 5$) in populations of the S536F (A) and +P518 (B) resistant mutants. Frequency of sensitives through time was marked by a rapid increase to approximately 0.5 and then a gradual increase to 1, showing more variability in S536F populations (A) than +P518 populations (B). Using this data, we calculated an effective selection coefficient (C) for the sensitive lineage relative to the resistant lineage (mean $\pm$ SE; $n = 5$) for each time frame in populations founded by S536F (white bars) and +P518 (gray bars).

because clonal interference allows strongly beneficial mutations to drive weakly beneficial mutations extinct. In our experiment, we found that a minority of sensitives that escaped genetic drift at the outset of the experiment were ultimately driven extinct, suggesting that the occurrence of supercompensatory mutations increasing fitness beyond the fitness of the current sensitive type prevented the fixation of sensitives (Fig. 1). The likelihood of extinction due to clonal interference was twice as high for sensitives in S536F as +P518, with $13/38 = 0.34$ losses of sensitives that escaped drift in S536F populations and $9/54 = 0.17$ losses of sensitives in +P518 populations, with marginal statistical significance ($\chi^2_1 = 3.77$, $P = 0.052$). However, we note the low power associated with test due to the small number of sensitives observed (41–55%); to achieve power of 80%, we would have needed to track 96 sensitives in each resistant background.

This intriguing difference suggests that the two resistant mutants differ in their ability to generate the supercompensatory mutations that are required to drive established sensitive lineages extinct. To test this hypothesis, we measured the fitness of a random sample of 42 of the 265 evolved populations of each resistant genotype in which the sensitives never achieved a detectable frequency (i.e., $>0.0005\%$). Consistent with our earlier results (i.e., Fig. 1), we found that selection for compensation was very efficient, as evolved populations fully recovered the large costs associated with resistance. However, the gains in fitness were asymmetric as predicted, with S536F compensating significantly more than +P518, with fitnesses of 1.19 ± 0.037 SE and 1.02 ± 0.059 SE, respectively (Fig. 4; Welch's t -test: $t_{48.5} = -2.35$, $P = 0.023$).

SPONTANEOUS REVERSION OF RESISTANCE

We assayed for spontaneous reversion events by tracking the loss of the resistant type in the absence of the introduced sensitive. We did not observe any loss of resistance due to spontaneous reversion mutations in the 265 populations of resistant bacteria that we tracked for 300 generations. However, this does not preclude the possible occurrence of true spontaneous reversions that did not reach detectable frequency.

Discussion

In this study, we experimentally mimicked a mutation-reversing antibiotic resistance in populations carrying costly resistance mutations and then tracked the evolutionary dynamics of the sensitive and resistant lineages to determine how drift and compensatory adaptation influence the likelihood of fixation of a reversion mutation. We deliberately designed two features of this experiment to make it a “best-case scenario” for the fixation of reversion mutations. First, the cost of resistance was very large, which should maximize the likelihood of reversion by minimizing the likelihood

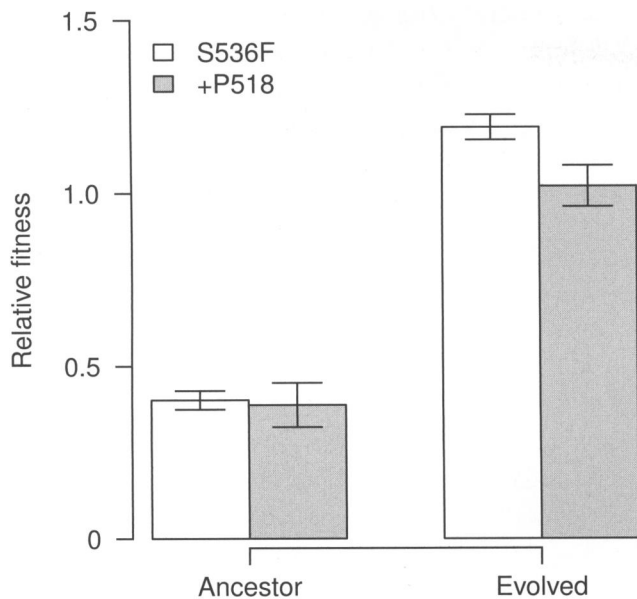


Figure 4. Evolvability of resistant mutants. This figure shows the relative fitness of the ancestral resistant mutants used to initiate the selection experiment (mean $\pm$ SE, $n = 5$) and the fitness of 42 evolved populations derived from each resistant mutant. Note that for this assay we only sampled evolved populations in which sensitive lineages were never detected.

that reversion mutations are lost due to genetic drift when they are initially rare. Second, we mimicked the appearance of reversion mutations before resistant populations had the opportunity to compensate for the cost of resistance, which should minimize the constraints on reversion imposed by clonal interference between reversion and compensatory mutations.

The majority of sensitives that we introduced were rapidly lost due to genetic drift, supporting a model in which the likelihood of a beneficial mutation (in this case, a reversion mutation) escaping drift is determined by the selection coefficient associated with the beneficial mutation (Wahl et al. 2002). Most sensitives that escaped stochastic loss at the start of the experiment were eventually fixed by natural selection, but both the time to fixation and the probability of fixation differed substantially between populations of different resistant mutants. The extent to which the resistant strains compensated for the cost of resistance played a key role in determining the dynamics of the sensitive fixation. This implies that the propensity to ameliorate the cost associated with resistance may be just as important as the magnitude of the cost itself in determining the dynamics of reversion.

MAINTENANCE OF RESISTANCE

When is resistance likely to revert? Given that the cost of resistance provides the incentive for reversion, intuition suggests that the cost of resistance should play an important role in determining the likelihood of reversion (Andersson and Hughes 2010). Inte-

grating the results of our experiment into the broader literature on compensatory adaptation suggests that, paradoxically, the cost of resistance may play only a minor role in the probability of reversion.

Our results agree with the earlier work on streptomycin resistance by Levin et al. (2000), which demonstrated that introduced sensitives were more likely to be lost stochastically if introduced into compensated instead of uncompensated resistant populations. Levin et al. (2000) demonstrated that compensated resistant strains could prevent the fixation of an introduced sensitive strain with higher competitive fitness, putatively by increasing the stochastic effect of random sampling. Contrary to our findings, they found that introduced sensitives rapidly rose to high frequency in uncompensated resistant populations without hindrance from novel compensatory mutations—this is likely because the sensitive strain was introduced at roughly four times the density of our experiments, making the changes in frequency more deterministic. One of the most striking results of our study is the finding that sensitives that escape drift always took a long time to achieve fixation, with an average fixation time about 10 times longer than the predictions of a simple model (Bell 2008) that ignores compensatory mutations (Figure 2). The cause of this discrepancy is clear: sensitives fixed very slowly because compensatory adaptation resulted in a rapid decline in the selection coefficient associated with the reversion mutation from 0.5 to 0.01 (Fig. 3), implying that both the +P518 and S536F mutants are able to easily acquire compensatory mutations with approximately equivalent benefits to reversion mutations. Is this surprising? The model of compensation that has emerged from mechanistic studies of compensatory adaptation (Maisnier-Patin et al. 2002; Kirschner and Gerhart 1998) is that compensatory mutations increase fitness by fully or partially recovering the biophysical defects associated with deleterious mutations, including resistance mutations. This biophysical model of compensation predicts that the benefits associated with compensatory mutations will likely covary with those of reversion mutations. The implication of this is that the likelihood of reversion could be largely independent of the cost of resistance; hopefully future studies will test this hypothesis directly.

COMPENSATORY ADAPTATION AND EVolvABILITY

Given that the rate and likelihood of reversion depends critically on the ability of resistant mutants to evolve compensatory adaptations, it is important to understand the factors that determine the evolvability of resistant mutants. It is often argued that evolvability is determined by broad features of protein and genome organization, such as robustness, redundancy, and modularity (Woods et al. 2011; Wagner 2008; Tokuriki and Tawfik 2009). Our results clearly show that subtle sequence variation can also have a profound impact on evolvability, at least in the short term (see

also MacLean et al. 2010a). Specifically, our results imply that +P518 has access to more compensatory mutations than S536F, whereas S536F has access to better compensatory mutations than +P518. A meta-analysis of compensatory adaptation shows that there are an average of approximately 11–12 compensatory mutations per resistance mutation (Poon et al. 2005), possibly because compensatory mutations tend to be found in close proximity to deleterious mutations in the three-dimensional structure of proteins (Kirschner and Gerhart 1998). Given that the opportunity for compensation is clearly limited, it is not surprising that resistance mutations differ in their evolvability. In the future, we plan to use a combination of detailed fitness measurements combined with whole-genome sequencing to understand the causes of this variation in evolvability.

Summary

This study challenges the long-held notion that it is the rarity of reversion mutations alone that impedes the reversion of antibiotic resistance and instead emphasizes the importance of the intrinsic evolvability of resistant populations for determining the dynamics of reversion. Future work needs to focus on understanding how the cost of resistance affects the likelihood of reversion and how the molecular mechanisms of resistance determine the ability to compensate for the cost of resistance. In a more applied context, we suggest that resistance management strategies should take information on both the short-term costs and long-term evolvability associated with resistance mechanisms into account (MacLean et al. 2010b).

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher’s website:

- Figure S1.** Competitive fitness measured using novel fluorometer-based assay and standard selective media plating.
- Figure S2.** Comparison of competitive fitness of strains measured versus GFP-PA01 and YFP-PA01.